Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Production AI systems engineer with 8+ years building financial data pipelines and LLM-powered applications at enterprise scale. Shipped Entrata's first production AI pipeline across 1.6M financial records (Python, PyTorch, OpenAI API) -- deep financial domain: HAP allocation, HUD payments, income eligibility, AMI-based pricing, tax credit calculations across 100K+ unit portfolios, zero compliance violations. Built agentic AI end-to-end: **AgentProbe** (3-agent LangGraph runtime, 5-metric LLM-as-judge eval engine, multi-provider LLM routing, RAG -- live at github.com/venkatalakshmiparimala/AgentProbe) and **Leasel** (8-agent HITL pipeline, 93.6% precision across 1,240 enterprise documents). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Production AI on Financial Data (1.6M records, compliance gating) | Multi-Agent Agentic Architecture (LangGraph, LangChain) | RAG Pipelines & Vector Search (ChromaDB) | LLM Evaluation & Model Assessment | Python · SQL · TypeScript | AWS (S3, EC2, RDS, Glue) · Azure | Financial Services & Compliance Engineering (HAP, HUD, LIHTC) | Enterprise Data Pipelines & ETL

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- **Designed and deployed an agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) -- operationalized across marketing and commercial survey workflows, delivering risk-aware compliance decisions with grounded, cited reasons.
- **Built a RAG retrieval layer** (ChromaDB vector engine, FAA Part 107 corpus) grounding every compliance check in source regulation -- eliminating AI hallucination on domain-specific rules. Engineered validation guardrails blocking non-compliant outputs before export.

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Managed end-to-end financial services AI and automation**: HAP allocation, rent pricing strategy (AMI-based pricing tiers, gross rent updates), tax credit calculations, HUD payments, income eligibility verification, ledger management, and recertification workflows -- analytical, compliance-gated financial data engineering across 100K+ unit portfolios; zero federal/state violations.
- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) -- LLM-powered automation replacing manual Excel operations; output validation, quality gates, and compliance controls at launch; zero incidents post-deployment.
- **Built and optimized enterprise data pipelines** for real-time financial operations -- optimized 200+ SQL queries (30% performance gain) via execution plan tuning, indexing, and partitioning; 99.5% uptime across 100K+ monthly active users over 6+ consecutive months.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines -- zero security incidents. Led Docker-based microservices redesign: 50% scalability improvement, 30% deployment time reduction.
- Eliminated 80% of manual operational workload via Python/Bash automation; conducted 300+ peer code reviews; mentored 5+ engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed faculty dashboards improving comprehension 40%. Mentored 600+ students in Python and database design.

PROJECTS

AgentProbe: Multi-Agent AI Platform & LLM Evaluation Engine Live

3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), tool-use patterns, and inter-agent state handoffs. CI-integrated 5-metric LLM-as-judge evaluation engine (factual accuracy, hallucination, retrieval, completeness, compliance) -- every commit auto-scored, regressions blocked before production. ChromaDB RAG pipeline with trace-level observability: per-run metrics, regression drift, and retrieval quality monitoring.

Python · LangGraph · LangChain · FastAPI · ChromaDB · React · Docker · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Leasel: Enterprise AI Document Intelligence & 8-Agent HITL Pipeline

Dual-mode enterprise AI platform: Document Intelligence (OCR ingestion, vector retrieval, 100% citation correctness, human-review gate for regulated decisions) + Support Investigation (8-agent LangGraph pipeline with conditional HITL escalation). 93.6% precision / 87.9% F1 across 10 compliance rules on 1,240 enterprise documents (5 US states + federal).

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Reduced manual clinical retrieval time ~60% by building an NLP platform (LLMs, BERT, embeddings) translating natural language queries (EN/ES) to FHIR APIs with HIPAA-compliant RBAC. Deployed on cloud infrastructure with Docker containerization.

Flask · React · LLMs · FHIR · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur 2017

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, R, Bash

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Agentic Architecture, HITL Design, LLM Evaluation, LLM-as-Judge, Prompt Engineering, RAG Pipelines, Vector Search, ChromaDB, Embeddings, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, GitHub Actions CI/CD, PostgreSQL, MySQL, ETL, Data Pipelines, Microservices, RabbitMQ

Financial Domain: HAP, HUD Payments, Income Eligibility, AMI Pricing, Tax Credit Calculations, LIHTC, HCV/Section 8 Compliance, Affordable Housing Regulations